King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 201 Major Exam II Term 251 (Version A)

Original exam date: 12 November 2025. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. If $z = \ln(2x^2 + y^2)$, then $x\dfrac{\partial z}{\partial x} + y\dfrac{\partial z}{\partial y} =$

(a) z^2

(b) 2

(c) 0

(d) $-z$

(e) z

2. If z is defined implicitly as a differentiable function of x and y by the equation

$$x \ln y + y^2 z + z^2 = 6,$$

then $\dfrac{\partial z}{\partial y}$ at the point $(4, 1, 2)$ is equal to

(a) $-\dfrac{8}{5}$

(b) $\dfrac{23}{19}$

(c) $\dfrac{9}{4}$

(d) $-\dfrac{2}{3}$

(e) $-\dfrac{9}{4}$

3. The maximum value of the directional derivative of $f(x, y) = y^2 - x\sqrt{y}$ at the point $(0, 1)$ is equal to

(a) 3

(b) 2

(c) $\sqrt{3}$

(d) 5

(e) $\sqrt{5}$

4. $\lim_{(x,y)\to(0,0)} \dfrac{3x^3}{2x^2+2y^2} =$

(a) $\dfrac{3}{4}$

(b) DNE

(c) 3

(d) 0

(e) $\dfrac{3}{2}$

5. The domain of the function $f(x,y) = \sqrt{16-4x^2-y^2}$ is the set of all points

(a) lying on or inside the ellipse $\dfrac{x^2}{16} + \dfrac{y^2}{2} = 1$

(b) lying on or inside the ellipse $\dfrac{x^2}{4} + \dfrac{y^2}{16} = 1$

(c) lying outside the ellipse $\dfrac{x^2}{4} + \dfrac{y^2}{16} = 1$

(d) lying on or inside the circle $x^2+y^2=16$

(e) lying on or outside the circle $x^2+y^2=16$

6. Which one of the following statements is FALSE about the plane

$$4x - y + 3z = 6 \ ?$$

(a) It passes through the point $(1,1,1)$

(b) It has a normal vector $\vec{n} = \langle 4,-1,3\rangle$

(c) It is parallel to the plane $8x-2y-6z=10$

(d) It intersects the x-axis at $\left(\frac{3}{2},0,0\right)$

(e) The point $(0,-6,0)$ lies in the plane

7. If $f(x,y,z) = x^3 - 4xy + 3y^3 + xy^2z^2$, then

$$f_{xyzz}(1,1,1) + f_{xzyz}(1,1,1) =$$

(a) 10

(b) 16

(c) 8

(d) 2

(e) 14

8. Consider the unit vector $u = \frac{1}{\sqrt{2}}(i + j)$ and the two functions

$$f(x,y) = \begin{cases} \dfrac{4xy}{x^2+y^2} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases} \qquad g(x,y) = \begin{cases} \dfrac{4xy}{x^2+y^2} & \text{if } (x,y) \neq (0,0) \\ 2 & \text{if } (x,y) = (0,0) \end{cases}$$

Which of the following is FALSE?

(a) f is not differentiable at $(0,0)$

(b) $D_u f(0,0)$ does not exist

(c) g is differentiable at $(0,0)$

(d) $D_u g(0,0) = 0$

(e) $f_x(0,0) = 0$

9. Let $z = x^2 + y^2$. If (x,y) changes from $(1,1)$ to $(1.2, 0.9)$, then using differentials, the change in z is approximately equal to

(a) 0.05

(b) -0.1

(c) 0.4

(d) 0.02

(e) 0.2

10. If $ax + by + cz = 8$ is an equation for the plane passing through the points $P(2,1,0)$ and $Q(0,3,4)$ and is perpendicular to the xz-plane, then $a + b + c =$

(a) 3

(b) 2

(c) 4

(d) 6

(e) 8

11. An equation for the level surface of $f(x,y,z) = 4x^2 + 4y^2 - z^2$ that passes through the point $(1,1,3)$ is given by

(a) $z^2 = 4x^2 + 4y^2 + 3$

(b) $z^2 = 4x^2 + 4y^2$

(c) $z^2 = x^2 + y^2 + 3$

(d) $z^2 = x^2 + y^2 + 9$

(e) $z^2 = 4x^2 + 4y^2 + 1$

12. If (a, b, c) is the point of intersection between the lines

$$L_1 : \ x = 1 + 2t, \ y = 1 - 3t, \ z = 2 - 5t$$

$$L_2 : \ x = 8 - 3s, \ y = -8 + 4s, \ z = -14 + 7s$$

then $abc =$

(a) -17

(b) 27

(c) 38

(d) -28

(e) 25

13. Which of the following statements is FALSE?

(a) The graph of $16x^2 - y + 16z^2 = 4$ is an elliptic paraboloid

(b) The graph of $16x^2 - y^2 - 16z^2 = 4$ is a hyperboloid of two sheets

(c) The graph of $16x^2 - y^2 + 16z^2 = 4$ is a hyperboloid of one sheet

(d) The graph of $16x^2 + y^2 + 16z^2 = 4$ is a cylinder

(e) The graph of $16x^2 - y^2 + 16z^2 = 0$ is an elliptic cone

14. An equation of the tangent plane to the surface $z = y\, e^{2xy}$ at the point $(0, 2, 2)$ is

(a) $8x + y - z = 0$

(b) $32x + y - z = 0$

(c) $y - z = 0$

(d) $x + y - z = 0$

(e) $18x + y - z = 0$

15. If $w = x^2 - 3y^2$, $x = s\cos t$, $y = s\sin t$, then $\dfrac{\partial w}{\partial t} =$

(a) $-6s^2 \sin(2t)$

(b) $-2s^2 \sin(2t)$

(c) $-4t^2 \cos(2t)$

(d) $-4s^2 \sin(2t)$

(e) $-3s^2 \cos(2t)$

Answer Key

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Q	Ans	Brief explanation
1	**(b)**	$xz_x + yz_y = \frac{4x^2+2y^2}{2x^2+y^2} = 2$.
2	**(a)**	$z_y = -F_y/F_z = -\frac{x/y+2yz}{y^2+2z} = -\frac{8}{5}$ at $(4,1,2)$.
3	**(e)**	$\nabla f(0,1) = \langle -1, 2\rangle$, so the maximum is $\|\nabla f\| = \sqrt{5}$.
4	**(d)**	Polar: $\frac{3x^3}{2(x^2+y^2)} = \frac{3}{2} r\cos^3\theta \to 0$.
5	**(b)**	$16 - 4x^2 - y^2 \geq 0 \iff \frac{x^2}{4} + \frac{y^2}{16} \leq 1$: on or inside that ellipse.
6	**(c)**	Normals $\langle 4, -1, 3\rangle$ and $\langle 8, -2, -6\rangle$ are not parallel, so that statement is false.
7	**(c)**	Both $f_{xyzz} = f_{xzyz} = 4y = 4$ at $(1,1,1)$, so the sum is 8.
8	**(c)**	Limit at origin does not exist, so g is not continuous and not differentiable: that statement is false.
9	**(e)**	$dz = 2x\,dx + 2y\,dy = 2(0.2) + 2(-0.1) = 0.2$.
10	**(d)**	Perpendicular to xz-plane gives $b = 0$; P gives $a = 4$, Q gives $c = 2$: sum 6.
11	**(e)**	$c = f(1,1,3) = -1$, so $4x^2 + 4y^2 - z^2 = -1$, i.e. $z^2 = 4x^2 + 4y^2 + 1$.
12	**(d)**	$t = -1,\ s = 3$ gives $(-1, 4, 7)$, so $abc = -28$.
13	**(d)**	$16x^2 + y^2 + 16z^2 = 4$ is an ellipsoid, not a cylinder.
14	**(a)**	$f_x = 8,\ f_y = 1$ at $(0,2)$: $z - 2 = 8x + (y-2)$, so $8x + y - z = 0$.
15	**(d)**	$w_t = 2x(-s\sin t) - 6y(s\cos t) = -8s^2 \sin t\cos t = -4s^2\sin 2t$.